

2026.3

对辐照度整体的建模做了大幅优化

原:

$$TI(t) = DNI(t) \cdot \cos(\text{入射角}(t)) + DHI(t) + GD(t)$$

现:

$$TI(t) = DNI(t) \cdot \cos(AOI) \cdot fIAM(AOI) + DHI(t) \cdot fIAM, diff \cdot ((1 + \cos\beta)/2) + GHI(t) \cdot \alpha_{eff}(t) \cdot ((1 - \cos\beta)/2) \cdot fIAM, diff$$

$$\text{其中: } \alpha_{eff}(t) = [kdiff(t) \cdot \alpha_{ws}(t) + (1 - kdiff(t)) \cdot \alpha_{bs}(\theta_{Sun}, t)] \cdot 0.98 \cdot flow_sun(\gamma)$$

以 Perez 模型为基础

-> e.g. $G_{sky} = DHI \cdot (F_{iso} + F_{circumsolar} + F_{horizon})$

`poa = pvlib.irradiance.get_total_irradiance( surface_tilt=beta, surface_azimuth=gamma, dni=dni, ghi=ghi, dhi=dhi, solar_zenith=theta_z, solar_azimuth=azimuth, albedo=, model='perez' )`

`G_beam = poa['poa_direct']`

`G_sky = poa['poa_sky_diffuse']`

`G_ground = poa['poa_ground_diffuse']`

已经做的 4 个优化:

1. 入射角修正

为了修正南极低太阳高度角下, 光伏板玻璃表面的角度反射损失, 本模型引入了入射角修正因子 ($fIAM$)。其中, 直射项修正参考 Sandia IAM 模型, 而天空散射项的修正因子 ($fIAM, diff$) 则参考 Martin and Ruiz 分析模型, 以确保在大入射角条件下有效利用辐照度计算的准确性。

$$IAM(\theta) = 1 - b_0 (1/\cos\theta - 1)$$

-> (\$1 2.4) Angular reflection losses at the front and rear module surface are accounted for with the **Sandia incident angle modifier (IAM) model** (King et al., 2004). An IAM profile for a c-Si module with non-antireflective coated glass was used for the rear side.

-> (\$2 2.2&3.1) One problem of the reference cell, however, is that reflections from its planar glass cover significantly reduce the measured effective irradiance at high angles of incidence. This effect can be modelled approximately using incident angle modifier (IAM) models. In Fig. 2 we show the result of fitting the 1-parameter **Martin and Ruiz IAM model** [23] to the reference cell angular response using a parameter $ar = 0.163$. This enables an angular correction to be applied, albeit with the introduction of significant uncertainty. We discuss the implications of the angular and spectral effects further in section 3.1.

In the first, we modify the reference cell albedometer measurement with an angular-response correction to reference cell irradiance measurements: where G'_{sky} and G'_{ground} are the incident-angle corrected irradiances measured, derived from the sky-facing and ground-facing reference cell measured irradiances G_{sky} and G_{ground} , respectively. $kdiff$ is the diffuse light fraction, $fIAM(\theta)$ is the incident angle modifier (IAM) function for direct light, and $fIAM, diff$ is the IAM for diffuse light. We used the Martin and Ruiz [23] IAM model with fit parameter $ar = 0.163$ and **$fIAM, diff = 0.986$ derived by assuming isotropic diffuse radiance**. As θ_{Sun} and $kdiff$ are not measured directly, they were inferred from calculation [29] and

estimation [30] respectively, using location, time and GHI as inputs. This method is relatively complex to implement and relies on estimation and assumptions.

2. 0.98 光谱错配

→(§1 3.3 fig12) The spectral nature of albedo is shown to have significant influence on bifacial photo-voltaic performance wherein backside spectral impact as high as 1.20 is observed for fixed-tilt systems above green vegetation and as low as 0.98 for systems above snow.

3. $\alpha_{\text{eff}}'(t)$

$k_{\text{diff}}(t) * \alpha_{\text{ws}}(t) + (1 - k_{\text{diff}}(t)) * \alpha_{\text{bs}}(\theta_{\text{Sun}}, t)$, 核心在于将到达地面的太阳辐射拆分为“直射”与“散射”两个物理分量, 并分别考虑地表对这两种不同光性质的反射特性。 α_{ws} 为 white sky albedo, while α_{bs} 为 black sky albedo, 这两个由 BRDF 卫星数据获得。

→(§1 2.3) In principle, the calculation of albedo from BRDF requires the 5- dimensional integration of the product of spectrally and angularly resolved incident radiance with the spectrally resolved BRDF. This is complex and requires more information than is typically available. For simplicity, it is convenient to separate into two components: White-sky albedo α_{ws} and black-sky albedo α_{bs} , representing diffuse and direct illumination components respectively. α_{ws} is calculated from a bihemispherical integral of the surface BRDF assuming an isotropic light source (diffuse sky). α_{bs} is calculated from a single hemispherical integral of the BRDF assuming that all incoming light is from a single direction (the Sun). α_{bs} is a function of solar zenith angle θ_{Sun} , whereas α_{ws} is independent of the Sun's position. The total albedo α estimated from a weighted sum of these sources:

$$\alpha = k_{\text{diff}} \alpha_{\text{ws}} + (1 - k_{\text{diff}}) \alpha_{\text{bs}}$$

where k_{diff} is the quotient of diffuse horizontal irradiance (DHI) divided by global horizontal irradiance (GHI). We used this approach to estimate effective albedo using the following procedure

4. flow_sun(y)

当太阳从 10° 降向地平线时, 太阳光在大气中穿行的路径极长。这导致水平面散射辐照度与水平面直射辐照度的比例 (DHI/GHI) 迅速增加。

BRDF 模型是基于经验核函数的数学拟合。当太阳处于极低角度时, 这些核函数会进入数学上的不稳定区, 导致计算出的 α_{bs} 数值可能出现物理上不可能的极值 (如大于 1.0 或剧烈震荡)

The daily horizontal diffuse to global fraction (K_d) on the clear days was less than 0.40, while on cloudy days K_d was greater than 0.95. Albedo increases with decreases in sun height on clear days, but only down to solar elevation angles of about 10° at which point the albedo decreases (Iqbal, 1983). This phenomenon occurs because the fraction of horizontal diffuse to horizontal beam irradiance rapidly increases as the solar elevation angle decreases from 10° toward the horizon. Therefore, we only show measurements when the sun is higher than 10° above the horizon in Fig. 4a, Fig. 4c, and Fig. 4e. (他没有记 10° 以下的数, 说明需要修正)

此系统还需做 4 个优化:

---> 雪损模型 (Snow Loss Model)

**[论文 3]: 单面组件在严寒环境下冬季雪损平均 33.12%。

--->电气失配与安装排布 (Portrait vs. Landscape)

面和组串(组串电流?)

--->阵列自遮挡 (Ground Self-Shading)

[论文 2] 光伏阵列会在地面投射长阴影。

--->双面或许更优